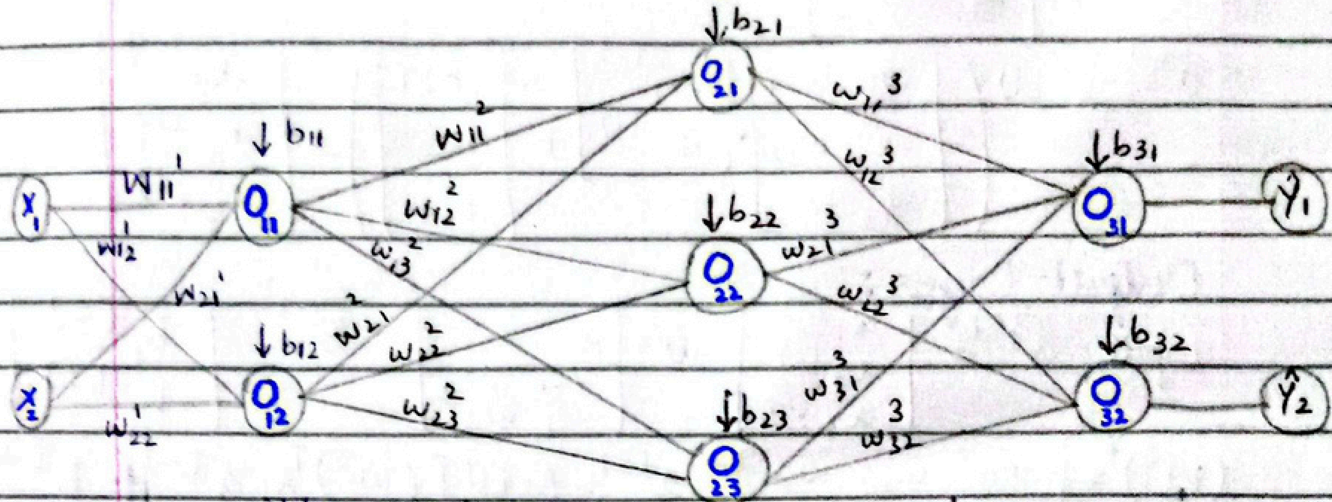
Take Home Quiz-4 Solution

Forward pass.



- Input layer (L_0): weights $= 2\times2 = 4$, + biases $= 4+2 = 6$
- hidden layer 1 (L_1): weight $= 2\times3 = 6$, + biases $= 6+3 = 9$
- hidden Layer 2 (L_2): weight $= 3\times2 = 6$, + biases $= 6+2 = 8$
- Output layer (L_3)

Total = 6+9+8 = 23 trainable params.

Hidden Layer 1:

$$\text{Input} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = a^{[0]} \qquad w^1 = \begin{bmatrix} w_{11} & w_{12} \\ w_{21} & w_{22} \end{bmatrix} \qquad b^{[1]} = \begin{bmatrix} b_{11}^{[1]} \\ b_{12}^{[1]} \end{bmatrix}$$

$$(w^1)^T = \begin{bmatrix} w_{11} & w_{21} \\ w_{12} & w_{22} \end{bmatrix}$$

$$a^{[1]} = \mathrm{ReLU}\left((W^1)^T \times a^{[0]} + b^{[1]}\right)$$

$$a^{[1]} = \mathrm{ReLU}\left(\begin{bmatrix} w_{11} & w_{21} \\ w_{12} & w_{22} \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} b_{11} \\ b_{12} \end{bmatrix}\right) = \begin{bmatrix} a_1^{[1]} \\ a_2^{[1]} \end{bmatrix}$$

Hidden Layer 2:

$$\text{Input} = a^{[1]} = \begin{bmatrix} a_1^{[1]} \\ a_2^{[1]} \end{bmatrix}, \; w^2 = \begin{bmatrix} w_{11} & w_{12} & w_{13} \\ w_{21} & w_{22} & w_{23} \end{bmatrix}, \; b^2 \begin{bmatrix} b_{21} \\ b_{22} \\ b_{23} \end{bmatrix}$$

$$(w^2)^T = \begin{bmatrix} w_{11} & w_{21} \\ w_{12} & w_{22} \\ w_{13} & w_{23} \end{bmatrix} \quad a^{[2]} = \text{ReLU}\left[(w^2)^T \times a^{[1]} + b^{[2]} \right)$$

$$a^{[2]} = \text{ReLU}\left(\begin{bmatrix} w_{11} & w_{21} \\ w_{12} & w_{22} \\ w_{13} & w_{23} \end{bmatrix} \cdot \begin{bmatrix} a_1^{[1]} \\ a_2^{[1]} \end{bmatrix} + \begin{bmatrix} b_{21} \\ b_{22} \\ b_{23} \end{bmatrix} \right) = \begin{bmatrix} a_1^{[2]} \\ a_2^{[2]} \\ a_3^{[2]} \end{bmatrix}$$

Output Layer:

$$\text{Input} = a^{[2]} = \begin{bmatrix} a_1^{[2]} \\ a_2^{[2]} \\ a_3^{[2]} \end{bmatrix}, \; w^3 = \begin{bmatrix} w_{11} & w_{12} \\ w_{21} & w_{22} \\ w_{31} & w_{32} \end{bmatrix}, \; b^{[3]} = \begin{bmatrix} b_1^{[3]} \\ b_2^{[3]} \end{bmatrix}$$

$$(w^3)^T = \begin{bmatrix} w_{11} & w_{21} & w_{31} \\ w_{12} & w_{22} & w_{32} \end{bmatrix} \quad a^{[3]} = \text{ReLU}\left[(w^3)^T \times a^{[2]} + b^{[3]} \right]$$

$$a^{[3]} = \text{ReLU}\left(\begin{bmatrix} w_{11} & w_{21} & w_{31} \\ w_{12} & w_{22} & w_{32} \end{bmatrix} \begin{bmatrix} a_1^{[2]} \\ a_2^{[2]} \\ a_3^{[2]} \end{bmatrix} + \begin{bmatrix} b_1^{[3]} \\ b_2^{[3]} \end{bmatrix} \right) = \begin{bmatrix} a_1^{[3]} \\ a_2^{[3]} \end{bmatrix}$$

As multiclass problem arises;

$$\text{Softmax}(a^{[3]}) = \begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \end{bmatrix}$$

$$\text{Softmax}\left((w^3)^T \cdot a^{[2]} + b^{[3]} \right)$$

$$\text{Softmax}\left((w^3)^T \cdot \left[\text{ReLU}\left((w^2)^T \cdot a^{[1]} + b^{[2]} \right) \right] + b^{[3]} \right)$$

$$\text{Softmax}\left((w^3)^T \left[\text{ReLU}\left((w^2)^T \cdot \left[\text{ReLU}\left((w^1)^T \cdot a^{[0]} + b^{[1]} \right) \right] + b^{(2)} + b^{[3]} \right) \right]$$

$$= \begin{bmatrix} a_1^{[3]} \\ a_2^{[3]} \end{bmatrix} = \begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \end{bmatrix}$$